

Ideas Parked — Not Yet Merged (v16)

Staging document; each entry marks its eventual destination

Two documents, read together: an external-style review of `PatternArithmetic.v15_21.tex` across simplicity, consistency, and reproducibility, and a response proposing a concrete two-paper split. Before parking either, the review’s checkable claims were verified directly against the current paper rather than taken on faith — every specific number, every claim about where a term is or isn’t used, checked and reported below. Nothing here is adopted; this is the full record for a future decision.

1 The review, verified

Every worked number checked, all exact matches. Sun-then-Wind $(-0.866, -0.500, 0.011)$, Wind-then-Sun $(0.278, -0.500, 0.820)$, the $0.6\angle 0^\circ/0.8\angle 120^\circ$ Mix result $r' \approx 0.721$ at 73.9° , the equator offset at 120° , and the enclosure freeze step counts (628 at $\delta = 0.02$, 1261 at $\delta = 0.01$) all appear in the current paper exactly as the review states them. Not approximately — checked character for character where the paper prints them.

“Axiom 7 reads as a second paper lodged inside the first” — checked, and true without qualification. Every occurrence of S_{atom} , $\mathcal{V}_{\text{atom}}$, γ_{atom} , $\mathcal{E}_{\text{atom}}$, α_{atom} , and σ_{atom} in the current paper — 22 occurrences total — falls strictly inside Axiom 7’s own subsection. Not one appears in Bundle, Mix, T , or offset’s own treatment anywhere else in the document. The review’s claim is not an impression; it is a count.

“White is used twice” — checked, and the paper already says so in its own words. Axiom 3 states, as its own equation: “White = saturation of the bowl, or a full-spectrum name.” Two senses, acknowledged in the text itself, exactly as the review reports — not a hidden overload the review discovered, but one the paper already names and moves past.

The “checked directly” precision claims — present, and genuinely without a reproducibility path. The inversion-error table (1.1×10^{-16} at several dimensions, 5.6×10^{-17} at others), the UnMix residual of 10^{-17} , and the “verified to six decimal places” line for the $\mathcal{E}_{\text{atom}}$ identity are all present in the current text, stated as results. No supplementary code, configuration file, or appendix exists anywhere in the document or its outputs. A reader has exactly what the review says: the printed number, and the formula it came from, with the orientation convention for higher-dimensional cases left to be reconstructed.

The technical consistency claims — checked, all hold. $A \uplus A \uplus A \uplus A \setminus A = \emptyset$ appears verbatim. Mix’s non-associativity under the cap is stated directly: “the cap makes the binary operator non-associative.” Both match the review’s description exactly.

What this verification adds. Not a correction — every claim checked came back accurate. What it adds is that the review’s recommendations now rest on confirmed facts about the current paper, not on an impression of an earlier or different one. The “cut Axiom 7” recommendation in particular is now a verified count (22 uses, 0 outside Axiom 7), not a stylistic judgment to weigh against a competing stylistic judgment.

2 The two-paper split, as proposed

The cut. Paper 1 (the typed algebra): the 4-tuple, Axiom 0, Bundle and Remove, Mix and the refusal across directions, Collapse and UnCollapse, T , offset, the sort table, and the UnMix argument. Its sentence: a combination rule should match the unit, and these verbs preserve different things. Paper 2 (assembly and the enclosure): the assembly signature, the energy, seats, the freeze, camps and bodies, and the printed six-unit replay. Its sentence: two units can be related without being fused, and seats left under that pull freeze into camps only at a stated threshold.

Where Axiom 7’s metrics and the costumes go: neither paper. Collapse needs only the counted list of vectors (N and the vectors themselves), not the seven derived statistics. The enclosure needs only r and, optionally, a resultant weight. The storm, forest, table-and-chair, and pathogen costumes restate $\uplus$ versus T and are required for neither paper’s own argument.

What must travel, specifically, or an idea breaks. Four items, stated as requirements rather than suggestions:

- Paper 2 needs roughly two pages of shared preamble from Paper 1 — the 4-tuple, the fact that nothing in $(r, \hat{n}, \chi, \tau)$ says where a unit sits, and the repulsion-free bracket $3 + \hat{n}_A \cdot \hat{n}_B \in [2, 4]$ — cited, not rebuilt.
- Paper 1 should state Assembly as a signature only: both units survive, the new object is a contact and an energy that asks for a seat. The full U_{AB} formula, which depends on d_{AB} , belongs in Paper 2, since the current text already says the verb is incomplete until the enclosure exists; leaving a half-defined U_{AB} in Paper 1 is named as the one genuine risk to the split.
- `chladni` stays in Paper 1. It is the evidence that Mix across directions is refused, not a result about plates, and the type table exists because of it.
- The higher-dimensional form of T stays in Paper 1, with offset’s plane transport alongside it. Neither depends on seats.

What the split costs, named directly rather than minimized. A reader of Paper 2 will not see why Mix is non-associative; a reader of Paper 1 will not see a camp freeze. This is called a loss of juxtaposition, not of either idea: each example already stands on its own printed numbers (verified above) without the other half present for rhetorical support.

The two wrong ways to make this cut, named explicitly as a warning. Splitting at Axiom 7’s position in the document, or moving Collapse to Paper 2 because it comes later in the text, separates a verb from the unit it acts on — a structural mistake, not a simplification. Splitting off related work or the costumes is a deletion, correctly identified as not the same operation as writing a second paper.

The condition for the split to work. Paper 1 ends by naming the seat it refuses to supply; Paper 2 begins from that refusal. The cut is by question, not by length.

3 Where this connects to the metrics already parked (ideas_v13–v15)

Not noted in either source document, since neither had the other’s context: the “cut Axiom 7’s metrics” recommendation and the mass/density/energy naming question already under discussion are the same material viewed from two directions. ideas_v13 asked which of S_{atom} , V_{atom} , γ_{atom} , $\mathcal{E}_{\text{atom}}$ earn a physics name; this review asks whether the whole catalogue earns a place in *this* paper at all, given that none of the four verbs actually reads from it. The D/N pair added to the Collapse section since (mass-number/ atomic-number) is unaffected either way: it already depends only on the counted list of vectors, which — per the split proposal above — is exactly what travels with Collapse regardless of where the seven-metric catalogue ends up. If the split is ever adopted, the open naming questions from ideas_v13–v15 would need to be re-homed wherever $\mathcal{E}_{\text{atom}}$ and its siblings land, or shelved as belonging to neither paper’s own argument, the same way the costumes are described here.

4 Status

Nothing in this entry is adopted. The review’s factual claims are now independently verified rather than merely asserted; the split proposal is recorded in full, including its own stated risk (a half-defined U_{AB} left in Paper 1) and its own stated conditions for success. The connection to ideas_v13–v15 is new, surfaced here for the first time, and is itself undecided.